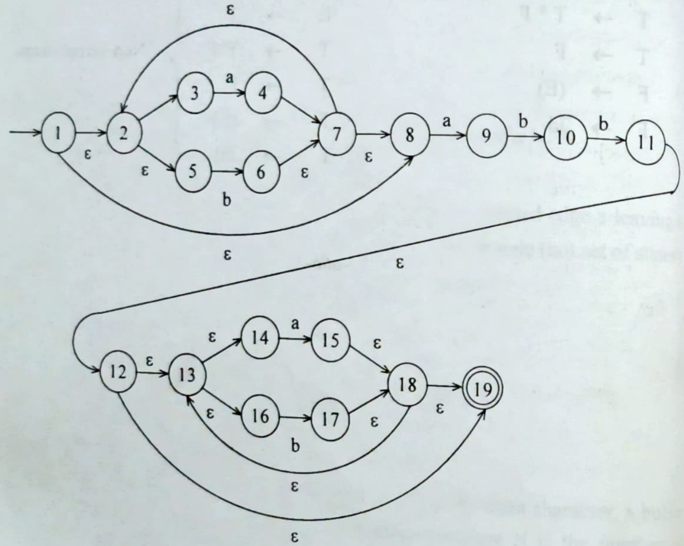
PART – B

SOLVED PROBLEMS

1. Convert the given regular expression (a|b)*abb(a|b)* into NFA using Thompson construction and then to minimized DFA. ***(APR/May 2010)***

Solution:

i) Convert the expression into NFA using Thompson construction.



ii) Convert the NFA into a DFA by eliminating ε-transitions.

Find the ε-closure of all the states

ε-closure(1) = {1,2,3,5,8}

ε-closure (2) = {2,3,5}

ε-closure (3) = {3}

ε-closure (4) = {2,3,4,5,7,8}

ε-closure (5) = {5}

ε-closure (6) = {2,3,5,6,7,8}

ε-closure (7) = {2,3,5,7,8}

ε-closure (8) = {8}

ε-closure (9) = {9}

ε-closure (10) = {10}

ε-closure (11) = {11,12,13,14,16,19}

ε-closure (12) = {12,13,14,16,19}

ε-closure (13) = {13,14,16}

ε-closure (14) = {14}

ε-closure (15) = {13,14,15,16,18,19}

ε-closure (16) = {16}

ε-closure (17) = {13,14,16,17,18,19}

ε-closure (18) = {13,14,16,18,19}

ε-closure (19) = {19}

Let the initial state be the ε - closure (initial state)

ε-closure (1) = {1,2,3,5,8} ....A

Move (A, a) = ε-closure ($\hat{\delta}$ (A, a))

= ε - closure (4, 9)

= {2,3,4,5,7,8,9} ....B

Move(A, b) = ε-closure (Move(A,b))

= ε-closure(6) = {2,3,5,6,7,8} C

Move (B, a) = ε-closure (Move (B, a))

= ε-closure (4, 9) = {2,3,4,5,7,8,9} ....B

Move (B, b) = ε-closure (Move(B, b))

= ε-closure (6, 10) = {2,3,5,6,7,8,10} ....D

Move (C, a) = ε-closure (Move (C, a))

= ε-closure (4, 9) ....B

Move (C, b) = ε-closure (Move (C, b))

= ε-closure (6) C

Move (D, a) = ε-closure (4, 9) ....B

Move (D, b) = ε-closure (4, 11)

= {2,3,4,5,7,8,11,12,13,14,16,19} ....E

Move (E, a) = ε-closure (4,9,15)

= {2,3,4,5,7,8,9,13,14,15,16,18,19} ....F

Move (E, b) = ε-closure (6,17)

= {2,3,5,6,7,8,13,14,16,17,18,19} ....G

Move (F, a) = ε-closure (4,9,15) ....F

Move (F, b) = ε-closure (6, 10, 17)

= {2,3,5,6,7,8,10,13,14,16,17,18,19} ....H

Move (G, a) = ε-closure (4,9,15) ....F

Move (G, b) = ε-closure (6,17) ....G

Move (H, a) = ε-closure (4,9,15) ....F

Move (H, b) = ε-closure (6,11,17)

= {2,3,5,6,7,8,11,12,13,14,16,17,18,19} I

Move (I, a) = ε-closure (4,9,15) ....F

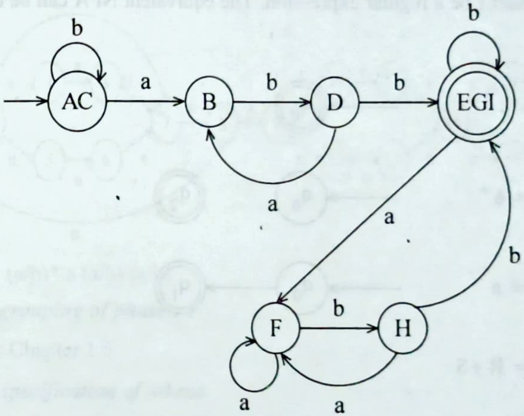
Move (I, b) = ε-closure (6, 17) ....G

Since there is no new states, stop with this.

	a	b
→ A	B	C
B	B	D
D	B	E
* E	F	G
* F	F	H
* G	F	G
* H	F	I
* I	F	G

After eliminating duplicate states, the minimal DFA is as follows.

	a	b
→ AC	B	AC
B	B	D
D	B	EGI
* EGI	F	EGI
F	F	H
H	F	EGI



2. *Explain in detail about the phases of compiler and translate the following statement, pos := init + rate * 60*

Refer Chapter 1.3

3. *Discuss any 4 compiler construction tools.*

Refer Chapter 1.6

4. *Write down the exclusive functions carried out by the analysis phases of any compiler.*

Refer Chapter 1.2

5. *Write briefly on the issues in input buffering and how they are handled.*

Refer Chapter 1.8, 1.8.1